

# SPECTRAVIDEO USERS GROUP

WELLINGTON

## THE LAST NEWSLETTER

Vol 4 - Number 6

APRIL 1989

EDITOR: John Hodgson

(00 64) 4 367550

.....

Meetings were held on the third Monday of the month at 7.30 pm in the Staff Room of Newlands Intermediate, Bracken Road, Newlands, Wellington. Attendance used to average 10-15 with ideas, information, problems and matters of interest being shared.

.....

### COMMITTEE MEMBERS

|                     |               |            |
|---------------------|---------------|------------|
| Chairman            | Jim Oliver    | ph 665 205 |
| Secretary/Treasurer | Ross Fletcher | ph 788 454 |
| Librarian           | Scotty Baxter | ph 784 619 |
| Public Domain BASIC | Jim Oliver    |            |
| Public Domain CP/M  | Ross Fletcher |            |

NEWSLETTER CONTRIBUTIONS TO: J. Hodgson, 38 Gloaming Hill  
Onepoto  
WELLINGTON

=====

### CONTENTS

|   |                                             |         |
|---|---------------------------------------------|---------|
| : | :                                           | :       |
| : | :                                           | :       |
| : | Guest Editorial .....                       | page 2  |
| : | :                                           | :       |
| : | Explaining the Artist Program .....         | page 3  |
| : | BRAIN STRAIN .....                          | page 6  |
| : | Art Program - Michael Warwick .....         | page 5  |
| : | User Notes - Fisk's Disk Utility .....      | page 7  |
| : | 800 Column Screen Print Dump .....          | page 9  |
| : | Public Domain CP/M .....                    | page 8  |
| : | The Basic "INSTP" Command .....             | page 10 |
| : | Explanations from Bryan Fisk .....          | page 11 |
| : | Catalogue of SVI Cards/Modem/Hardware ..... | page 13 |
| : | THE END From Your Editor .....              | page 14 |
| : | :                                           | :       |
| : | :                                           | :       |
| : | :                                           | :       |

Guest Editorial - from your Chairman

Your Committee has decided to officially wind up the Wellington Spectravideo Users Group.

This decision has been reached only after careful consideration of any advantages of remaining active.

Our current editor is unable to carry on due to personal commitments which leave him little time for preparing our newsletter. This is a time consuming job and can be very frustrating in the absence of material. There appears to be no-one able/willing to take this responsibility on.

With a few notable exceptions there has been little input from those whom we are serving. As usual in these cases the main burden of work and copy material has been left to a few people.

Our dwindling membership lists further prove the necessity of winding up the group.

However lets be positive, life goes on, and a Wellington member suggested we join up with the Osborne Users Group. Some members have been along to a few of their meetings and find that their group has an extensive library of both CP/M and MS-DOS discs (about 500 in all), they have a newsletter and offer access to FOGHORN and FOGLIGHT. These last two are really excellent magazines produced in the USA by the OSBORNE group called First Osborne Group. They are for CP/M and MS-DOS users respectively.

They have a group in Auckland called the NZ Osborne Users Group and it may suit out of town members to join them.

What we are doing is negotiating a hopefully free upgrade to Osborne User Group membership for our current members. If you are interested in having the remainder of your CURRENT PAID UP MEMBERSHIP transferred then write immediately to the Treasurer, Ross Fletcher, see front cover.

The Osborne PD library will be available to anyone who joins up. Since this library is so vast detailed lists are not available, however they do have an excellent disc based data base which can be searched on any keyword and will show which disc has the required files. This takes up 2 discs for both CP/M and MS-DOS databases. Send your discs after joining up to me and for our normal charge (\$3.00 per disc + \$2.50 postage and handling) I will copy these for you. The data base comes with built in instructions.

I am personally very impressed with the group and their well run library and will not hesitate to join up and subscribe to the FOGHORN newsletter at about US \$25 per year for 64 pages each month. There's quite a bit on Basic in this magazine as well, and it's MBASIC which is easily converted to SV basic.

Well there it is. We think we've done quite well since June 1984,

there are not too many other groups who have outdone us over such a long time. Remember this : the Osborne group have a subscription list of 17,000 members ! We have about 50.

Note that our post box at Newlands is still active and any correspondence can be sent there.

Our PD library will be incorporated into the Osborne Users Group in Lower Hutt (known as Central Region Osborne Group or CROG). I will still be operating this library and discs can be got through me for the normal charges as above. However since our post box is operative for only a short while you would be best to address these requests directly to me.

Jim Oliver

# Explaining the Artist Program By Michael Warwick

This is a guide to the artist program to get you started. I will go into some detail on most of the comands, but first I suggest you (if you haven't already) type in the artist program.

This program was designed to make the best possible use of the keypad on the SVI-328, the only keys which haven't been used are the [select] and [print] keys which can be used by a machine code dump as the one already in the program is rather slow (up to 10 mins for a printout!!).

When you run the art program you will find it takes a little while before it actually starts. This is because it is loading the font into an array, the font is the one you are currently using so special fonts can be used. If you have a font editor just load the font you want to use, and then exit the font editor & run the art program. Any notes relating to the printer are for my Panasonic KX-P1080 which uses the Epson 'standard' and so should run on most other printers.

The program first starts (after loading the font) you will see a small cross in the centre of the screen, in the middle of this cross is a hole and this is where every-thing happens. Press the spacebar. The dot in the middle should turn black. Now move the cross away using the cursor keys and admire that dot. now you can draw things! if you put a dot in the wrong place it can be removed by simply pressing 'X'. Now if you want to add colour then press '+' and the cross should change to green (unless you are like me and have an amber monitor or green screen), pressing '-' will return your colour to black. If you press the '+' key enough times the cross will go through all the colours to white then to transparent and black again (if you press '-' the same thing happens in the reverse order). If you find the cross is moving to slow then press ',' and it will go twice as fast. pressing '.' again will make the cross go faster, and faster, until it reaches maximum speed, when it will go back to its old pace.

Now we get into some of the exciting graphics offered by the numeric keys, but first we will clear the screen by pressing '/'. Just one more thing, press the '.' key. This seems to do the same thing as the space, but really it is putting that point into memory for later use. We will start with key '9' and work down to Key '1'.

Key '9': this key paints in an area in the current cross colour. the border of the shape to be filled should be the same colour of the cross or it will just fill the entire screen.

Key '8': draws a box with one corner at the point in memory (where you last pressed '.') and the other at the present cross position. The box is in the present cursor colour.

Key '7': same as '8' except this command will fill the box (even if there is something in the area where the box is drawn).

Key '6': draws a line from point in memory to the present cross position.

Key '5': draws a circle, with the point in memory as the centre and with the present cross position as the outside.

Key '4': draws a circle as with '5' but then proceeds to paint in the circle as with '9'. Unlike the fill done with '7' this paint will not fill in an area inside the circle which has a border the same colour as the cross colour.

Key '3': this function is the very reason I sent this program in for in the first place. Pressing this key disables all of the other functions and puts you into 'type mode'. This means that you can type to the screen in much the same way as I am writing to Wordstar now. Although you don't have all the fancy editing functions you can print in any type-face you can print on the 40 col screen, and a multiple of different sizes. Once you have pressed '3' you can type any thing and it will appear onto the screen. Pressing the up arrow key will make the typing twice as large, and pressing it again will make it even bigger - the only limit is when the size number gets too large and causes a 'overflow' error. When you want the lettering to return to its normal or default size press the down arrow. If you get to the edge of the screen it will automatically go down to the next line.

Key '2': this key photographs a 10x10 area of the screen. The picture that has been taken is printed up in the top right hand corner of the screen.

Key '1': this key prints the photograph taken by '2' onto the screen. If you haven't taken a photo yet then you will get a error message and whatever was on the screen will be lost, never to be seen again! (unless you have saved it of course!). for those of you who are familiar with basic key '2' uses the get command and key '1' uses the put command. This means that it won't always work and quite often funny things can happen, if they do try pressing '1' again, maybe even 2 times. If this



doesn't work then you might have to fix it up using the space bar and '\*' key.

Pressing 'S' saves your work of art to disk, and pressing 'L' loads a screen from disk. Make sure when you load that you have the right disk in the drive, also be careful when saving that you are not writing over another picture / program of the same name.

Printing your master piece to paper is a slow business so if you can I advise that you use a machine code dump which uses the print key and/or the select key. If you can't do that then there is one built in. Press ^P (CTRL and 'P' at the same time) and then '1' for a small picture or '2' for double size and leave the computer to print it out. The cross shows you where the computer is at and the printing happens each time the cross reaches the edge of the screen.

Remember that colours may begin to 'run' because of the way the screen is stored in memory. You can only have two colours in a 8 pixel long line to put it simply although most of the time this does not matter. If you want more detail try reading 'Peeks and Pokes for the Spectravideo' by Deane Whitmore.

I have a pattern fill which fits nicely on the end of the artist program. It changes the way keys 1,2,4 & 9 work. The pattern fill does not work in the same way as the paint command and if anyone can find a better way to get the same effect please tell me.

To implement the pattern fill add these lines;

```
1000 FOR I = SX TO 0 STEP -1:IF POINT (I,SY)<>A THEN S=I+1 ELSE
--NEXT:S=0
1010 FOR I = SX TO 255:IF POINT (I,SY)<>A THEN E=I-1:RETURN ELSE
--NEXT
1020 E=255:RETURN
1500 A=POINT (SX,SY):GOSUB 1000
1510 FOR I=S TO E:FOR J=SY TO 191
1520 IF POINT (I,J)=A THEN GOSUB 2000:NEXT
1530 FOR J = SY-1 TO 0 STEP -1:IF POINT (I,J)=A THEN GOSUB
--2000:NEXT ELSE J=-1:NEXT J
1540 NEXT I:RETURN
2000 IF C(IMOD8,JMOD8)=1 THEN PSET (I,J),C ELSE PRESET (I,J)
2010 RETURN
```

Then change these lines;

```
20 DIM C(10,10)... BECOMES DIM C(8,8)...
130 ...VAL(A$)> 8 THEN160 BECOMES ...VAL(A$)> 9 THEN170
140 ADD ',1500' TO THE END DELETE 160
580 BECOMES ;580 FOR I=0 TO 7:FOR J=0 TO 7:IF POINT(SX+I,SY+J)=C
--THEN C(I,J)=1 ELSE C(I,J)=0
ADD THIS LINE;585 NEXT J,I:RETURN
590 BECOMES ;590 FOR I=0 TO 7:FOR J=0 TO 7:IF C(I,J)=1 THEN
--PSET(SX+I,SY+J),C
ADD THIS LINE;595 NEXT J,I:RETURN
```

And now you have a pattern fill.

I got the basic idea from an artist program written for the Amstrad (the programmer got 10 Pounds!).

The main problem with this fill is that when you fill most shapes some areas don't get filled, if anyone finds a solution to this problem (besides moving the cross to the problem area and the filling it!) please tell me. I would be most relieved. This fill fills until it reaches a pixel of a different colour to the point where the cross is unlike the paint command which fills till it reaches a pixel of the same colour to the point. The fill is activated by pressing '9' as before. Now '2' only takes a picture of an 8x8 grid and it only recognises dots that are the same colour as the cross. The picture you take with '2' is the pattern used in the pattern fill, and the pattern defaults to a white square.

Key '1' now only puts a 8x8 square to the screen. The pattern now has a white background and the pattern is printed in the colour of the cross.

Now the rest is up to you and your creativity! I'm sure you will get a lot of fun and interest out of it.

now go to it!

....000000000....

### BRAIN TEASER No.2 ANSWER

I found this simple problem quite difficult at first. Once I got the hang of it however I realised that there were several different ways of tackling the program. The one I've chosen to give you below is maybe not the fastest or the most elegant but it is the easiest to understand.

Firstly you need to know for speeds sake that our computer carries out arithmetic operations fastest when they are expressed in STRING form. This is convenient because we need to add a "3" to the centre of the top number.

```
10 REM No.11 'Brainteasers' Encyclopedia
20 TIME=0:B$="12456789":T=VAL(B$)
30 B$=RIGHT$(STR$(T),8)
40 TN$=LEFT$(B$,2)+"3"+MID$(B$,3,2):BN$=RIGHT$(B$,4)
50 IF VAL(TN$)=2*VAL(BN$)THEN PRINTTN$,BN$:PRINT(TIME/50)
-- ELSE T=T+1:GOTO 30
```

What we do here is as follows : start off with a number in B\$, note there is no "3". The reason for this is we want the 3 to be in the middle top number. Next we split out the two numbers and add in the "3". We then check to see if the top is twice the bottom and if it is print out the answer and the time taken to get there. If the numbers don't work out then increment T and go back again.

There are ways of checking to see if the number I contains only the numbers we want, i.e no zeros or repeated numbers. However I found that by doing this the program is slowed down further, not speeded up. There are ways of speeding it up ; since (HINT, HINT) the top number must start with "15.."

Have you found a quicker way ? If so let us know.

Jim Oliver

### BRAIN TEASER No. 3

I can only think that the reason we didn't get any replies to Brain Teaser No.2 was that it was too hard. Well here's a much easier one for you. And this time lets have a few replies from the budding programmers out there.

Here's four numbers :  
1731  
5363  
7179  
9903

Your job is to write a program to find the LARGEST number that will divide into each of these numbers and give the SAME REMAINDER.

Hint, it won't be any larger than 1731 !

### Notes for Users of Bryan Fisk's Disc Utility by Jim Oliver

I regularly back up the directories of our PD discs whenever I add or subtract new files. Sometimes I copy these discs for members using the CP/M COPY.COM file. I have found that some discs which have had the "directory saved as backup" using Disc Utility do not perform quite as expected.

The copying procedure works OK up to track 20 sector 7, then things go wrong and COPY will not go any further due to BAD SECTOR ! If this disc is checked using the Basic program called DSKTST, the disc checks out OK, but bombs with CP/M's FBAD.COM at group 49H or sector 8. Very strange. (Did you know that FBAD.COM will check a basic disc ?) I can't produce this failure every time but it happens too often to ignore. Maybe this has something to do with DSKOS not writing EXACTLY where it should.

Another problem is this : if you select option 6 from the main menu (save / load directory) and using a disc which has not as yet been backed up, then select the LOAD option in place of the SAVE option you will of course load a blank directory (none has been saved yet) on top of the current directory tracks. The result of this is the same as a format ! You completely ruin your directory. Maybe Bryan can fix this little beauty.

See Brian's article further on .... Ed

PUBLIC DOMAIN CP/M INFO

There are a few CP/M files that I use often. I thought it may be an idea to share these with some of you who may have just started with CP/M and could benefit from some basic info. (I certainly, could have)

FBAD56.COM

This COM file is used to check a newly formatted disc for bad sectors or bad system or directory tracks. Any bad sectors found will be "locked out" for general use by assigning them the file name [UNUSED BAD]. As far as CP/M is concerned these areas are now "in use" and will not be used again. This avoids the objectionable BAD SECTOR error (usually in the middle of a large Wordstar or dBase file!)

FBAD56.COM is available in several earlier versions but I always use V.5.6. The syntax for using this file is "FBAD56 D:" where D: is the drive you are testing. I now run this file on all formatted discs including BASIC ones. (However if an error is found on your BASIC disc some funny data is written to track 3 ' in which case you must re format.)

RECLAIM.COM

This is the best of the UNERASE or RECLAIM programs around. Simply type RECLAIM D: where D: is the drive name (if not the default drive) and you are presented with a menu. At the bottom of this is the first of the ERASED files on the disc. You may move to the next ERASED file by pressing <RETURN>. When you get to the file you want to RECLAIM you press "R" then "W" then "Q" (to Quit) What you must realise when using RECLAIM is that that when you ERASE a file all that happens is that the first byte of this files directory entry on the disc is altered from zero to E5 Hex. The actual data is still all there on the disc, PROVIDING you haven't written any new data to it. Even if you have used the disc briefly it's still worth a go for a valuable file. If you find there are heaps of ERASED files on the disc that you are never going to use again it is well worth running the next file; SORTDIR.COM.

SORTDIR.COM

This program checks the directory tracks for ERASED files, then permanently removes these AND rearranges the directory so that it reads alphabetically. I have found this file invaluable for cleaning up Wordstar data discs which after a bit of use can be a real mess. Yopu cannot use RECLAIM.COM after SORTDIR.COM.

DU-VB6.COM and NSWP207.COM

I use both of these files more than any others. NSWP207.COM is version 2.07 of New Sweep and I now have a 12 page DOC file on disc for anyone who is interested. ANYONE who uses CP/M must have at least one version of NSWP for coping, printing, deleting, squeezing (& unsqueezing), renaming, and setting files status.



DU-V86 is a disc utility which will do virtually ANYTHING you may wish to do to a disc by operating directly on the disc itself. So you have to have a certain minimum amount of knowledge before you use this one. There is a good DOC file available for this one as well.

### SVDO.COM

This one is a small (8K) Wordstar NON-DOCUMENT ONLY CP/M editor. This means you can use it for writing letters but it is normally used for writing programs or other files because it does not support word wrap, reformatting or justifying. However it is very fast and does enable you to use nearly all the RAM due to it's small size. There is a 24K DOC file available !! If you want to use MBASIC or SVMBASIC then this editor is MUCH easier to use than the one supported by MBASIC.

**WARNING:** There are many different versions of this file and nearly all of them have been patched in some way to run on non SV systems. This file is called all sorts of names depending on which computer it is run. However we have the SV version which can be installed to suit your individual requirements.

### The "INSTR" Command

Have you ever used the command INSTR available in SV BASIC ? You should have since it is a very powerful and fast way of testing BASIC strings for certain conditions. The syntax used is as follows :

A=INSTR(I,X\$,Y\$)

The variable A returns a number which is the first occurrence of the string Y\$ in X\$. The optional I is the starting point for search in X\$ and must be less than 255 since any string <255 characters.

For example if X\$ = "ABCDEFGHJIMIJKLMNO" and I was trying to find my name in there (Y\$="JIM") then A would return 9. Or if I wasn't sure if my name was in the string or not I could say PRINT MID\$(X\$,INSTR(X\$,"JIM"),3) and if it was there then "JIM" would be printed out.

The INSTR command comes into it's own for use when searching data files for specific information. Data files consisting of records of known length and fields can be searched with speed and ease.

Directing a program to certain lines after inputting from the keyboard is very conveniently done using INSTR. For example

```
10 CM$="AaBbCcXx"
20 PRINT "Please type a letter (A --> C ; X to exit)"
30 A$=INPUT$(1)
40 ON INSTR(CM$,A$) GOTO 50,50,50,50,50,50,70,70
45 BEEP: ? "YOU CAN'T FOLLOW INSTRUCTIONS"
50 PRINT"You typed ";A$
60 GOTO 30
70 CLS:PRINT"END"
```

Note that the use of upper or lower case is allowed by the specification in CM\$. The line of 50's can be ANY line number depending on what you want to do.

Recently I wished to test a set of random numbers to for duplication of any numbers as generated by the program. This can be done in several ways but INSTR was the best. If in the above example A\$ is jumbled up in any way then line 120 is "run" on it the answer will always be the same PROVIDED all the numbers are unique. Try it with any number duplicated. Lines 50 - 100 give a random set of numbers BUT with each number used only once.

```
10 INPUT "Number of variables";L:L=L-1:REM Select 9 in this case.
20 IF L=0 THEN BEEP:GOTO 10:REM see notes at end for other no's.
30 DIM N(L):X=PND(-TIME):A$="123456789"
40 INPUT "No. of cycles";C
50 FOR Q=1 TO C
60 FOR J=0 TO L:N(J)=J:NEXT
70 FOR I=0 TO L:J=INT(RND(1)*(L+1))
80 IF N(I)=0 THEN N(I)=J:'for no zero
90 SWAP N(I),N(J)
100 NEXT
110 FOR I=0 TO L:B$=B$+STR$(N(I)):NEXT:PRINTB$;" ";
```

```
120 FOR V=1 TO L+1:S=INSTR(B$,MID$(A$,V,1)):SUM=SUM+S:NEXT:PRINT SUM
130 SUM=0:B$=""
140 NEXT Q
```

There is a way of generating the A\$ string automatically without having to define it first as I have done above, can you work it out ?

Jim Oliver

To Whom It May Concern,

from Bryan Fisk

Sorry about the time taken to investigate/reply to the problem with copying disk via CP/M COPY.COM, where the directory had been saved using disk.utl. What I seem to have found out is that even though the COPY program gives a fatal read error, when you repy with an I (for Ignore) the copy program writes the 'unreadable' sector anyway. When sector 8 of track 20 is examined it looks just like the one on the original disk! And the destination disk had been CP/M FORMAted before the copy!!

If you could confirm this independantly - whenever you get the fatal read error, (I)gnore it.

Also I have made some changes to disk.utl to solve the problem of overwritting the directory without having saved a directory on that disk first; now a warning is given when the program encounters a date/time stamp that it considers to be blank (and anyway when you 'load' a directory, you should check that the date/time stamp makes sense). Also now date/time fields MUST be entered otherwise an error is displayed. And I have changed the save/load directory to read/write 256 byte sectors instead of 255 byte sectors.

You should also find on the disk, a program called 'dir.chk'. This is a simple program to check that all the tracks that the File Allocation Table say are inuse are actually inuse by a file. Used mainly by me, when I have been playing around with the directory sectors and want to know if I had stuffed up any of the file info.

So what you should find on this disk is:

1. dir .chk - directory check program
2. disk .alt - a 'mergable' file of the changes to disk.utl
3. disk .dev - the development version of disk.utl (NOTE this program cannot now be run ,paradoxically, without being condensed via disk.utl i.e. for debugging purpose only)
4. disk .utl - the runnable version of disk.dev (already has the changes in disk.alt made to it)
5. RUN .ME - the program that displays this file
6. RUNME .DAT - this file

Note : Number 3 above is too large to fit into the newsletter and is best got from jim Oliver from the PD library. Alternatively merge the file which follows with disk.ult and you will then have the corrected version.

Bryan Fisk (disk.alt)

```

27 NO=0: IF NO<>0 THEN A=NO:POKE 32955',18:GOTO 190
40 PRINT"B          DISK UTILITIES   Version 1.02"
2990 POKE 32955',18:RUN
3700 POKE 32955',18:RUN
6140 FIELD ,128 AS A$, 128 AS B$
6159 PRINT
6160 IF MID$(Z$,1,20)=STRING$(20," ")
    OR MID$(Z$,1,20)=STRING$(20,255)
    OR MID$(Z$,1,20)=STRING$(20,229)
    THEN PRINT CHR$(27)"p""WARNING"CHR$(27)"q"
    " NO DATE/TIME STAMP 'CONTINUING' MAY DESTROY
    YOUR DIRECTORY!!" ELSE PRINT"LAST UPDATED ON ";MID$(Z$,1,20)
6220 ' LSET A$=Z$
6260 'LSET A$=Z$
6300 GOTO 1005
6340 FIELD ,128 AS A$, 128 AS B$
6354 IF D$="" THEN PRINT "TODAY'S DATE MUST BE ENTERED": GOTO 6352
6355 PRINT
6356 INPUT"CURRENT TIME          ";T$
6357 IF T$="" THEN PRINT "CURRENT TIME MUST BE ENTERED":GOTO 6356
6400 ' LSET A$=Z$
6440 'LSET A$=Z$
    
```

# IMPAK COMPUTING

P. O. Box 3153, Christchurch 8001, New Zealand. Telephone: (03) 656-664

## SVI HARDWARE:

- GO-101: PRINTER INTERFACE (Plugs straight into the back of the SVI-318/328). Cost: \$69.95
- SVI/CAB: SVI Printer Cable. Plugs into GO-101. Cost: \$24.95
- SVI-805: \*RS232 Interface Card. Cost: \$155.00
- SVI-806: \*80 COLUMN Card. Cost: \$195.00
- SVI-602: \* Single Slot Expander. Cost: \$45.00
- SVI-JOY: SVI Quickshot Joystick. \$14.95
- DE-P01: 360K Double-Sided 5.25" Disk Drive for the SVI. Cost: \$189.00

\* = Imported from US. Availability: 6/7 weeks.

GST & \$2.00 P&P must be added per order.

Unmarked (without the "\*" symbol) products are available within 24 hours.

MODEMS FOR THE SVI CAN BE FOUND IN CATALOGUE UNDER ~~PC-340~~ PC-340 & PC-34E ON PAGE # 6. SVI SOFTWARE SUPPLIED FREE OF CHARGE!



Well that's it folks. Many thanks to those who have supported this publication and the few stalwarts behind the scenes who have kept the club going this long. There have been some interesting people over the years and some great contributions. This is not meant to be a roll of honour but it is appropriate to pick one or two names out. Going way back to beginnings the efforts of Don Stanley will be remembered by many for his energy and dedication to the early years of the club, delving into the bowels of the SpectraVideo and contributing much of the knowledge documented in the newsletters; Ross Wakelin for his technical expertise in converting the hardware to run double sided 80 track drives in the B slot among other contributions on the CP/M operating side; Gary Clark for his efforts on Basic programming performed all those years without disk drives, Colin Chin and latterly Ross Fletcher for their contributions as Treasurer, Steve Hanson as long serving Editor of the Newsletter; Scotty Baxter for maintaining the Access Radio transmissions until their demise and latterly running the Club Library; and for me an outstanding contribution from Jim Oliver who has done perhaps more than anyone to keep the Club going over the last two years or so.

I've enjoyed it and I hope you have also. Thanks again for the support and good computing.

PS.

The 1985 NZ Technology Conference

voted the SpectraVideo 318/328

Personal Computer as

TOP PC OF THE YEAR

IN THE HOME AND SMALL

BUSINESS CATEGORY"